LIFTING A VALUATION RING THROUGH A FLAT MORPHISM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the flat going-down argument written out in full; the hypothesis that the closed point lies in the image of |X| -> |Y| is used exactly once, to place the point u.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Description of points of a fibre product of algebraic spaces. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian, used at proof lines 8977-8980. Lifts the closed point of Spec(A) to a point of X_A.
- Flatness is stable under base change. Lemma lemma-base-change-flat, used at proof lines 8977-8978. Keeps flatness of X_A -> Spec(A).
- The going-down property of flat local ring maps: primes lying over the generic point. Algebra, Lemma algebra-lemma-ff-rings, used at proof lines 8983-8987. Produces the prime q of O_{U,u} lying over (0) in A.
- Existence of a valuation ring dominating a local domain inside its fraction field. Algebra, Lemma algebra-lemma-dominate, used at proof lines 8988-8990. Produces A' contained in kappa(q) dominating B/q.
- Extension of valuation rings. More on Algebra, Definition more-algebra-definition-extension-valuation-rings, used at proof lines 8971-8973. Names the resulting relation between A and A' in the statement.

Source reference. Original lemma 00IA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 672–672. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 652–652. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03MO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4756–4756. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 0ASG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13579–13579. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03H4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4553–4553. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a flat morphism of algebraic spaces over S. Let $\mathrm{Spec}(A) \to Y$ be a morphism where A is a valuation ring. If the closed point of $\mathrm{Spec}(A)$ maps to a point of $|Y|$ in the image of $|X| \to |Y|$, then there exists a commutative diagram*

$$\begin{array}{ccc} \mathrm{Spec}(A') & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

where $A \to A'$ is an extension of valuation rings (More on Algebra, Definition 0ASG).

Proof. The base change $X_A \to \mathrm{Spec}(A)$ is flat (Lemma 03MO) and the closed point of $\mathrm{Spec}(A)$ is in the image of $|X_A| \to |\mathrm{Spec}(A)|$ (Properties of Spaces, Lemma 03H4). Thus we may assume $Y = \mathrm{Spec}(A)$. Let $U \to X$ be a surjective étale morphism where U is a scheme. Let $u \in U$ map to the closed point of $\mathrm{Spec}(A)$. Consider the flat local ring map $A \to B = \mathcal{O}_{U,u}$. By Algebra, Lemma 00HQ there exists a prime ideal $\mathfrak{q} \subset B$ such that $\mathfrak{q}$ lies over $(0) \subset A$. By Algebra, Lemma 00IA we can find a valuation ring $A' \subset \kappa(\mathfrak{q})$ dominating $B/\mathfrak{q}$. The induced morphism $\mathrm{Spec}(A') \to U \to X$ is a solution to the problem posed by the lemma. □